

main.py > ...

```
1  import pandas as pd
2  import matplotlib.pyplot as plt
3  import seaborn as sns
4
5  data = {
6      'Customer_ID': [1,2,3,4,5,6,7,8,9,10],
7      'Age': [22,25,35,45,29,31,40,50,27,38],
8      'Gender': ['F','M','F','M','F','M','F','M','F','M'],
9      'Marketing_Spend': [1000,1500,2000,2500,1800,2200,3000,3500,1700,2600],
10     'Sales': [5000,6000,9000,11000,7500,9500,13000,15000,7000,12000],
11     'Converted': [1,1,1,1,0,1,1,1,0,1]
12 }
13
14 df = pd.DataFrame(data)
15
16 print("Marketing Dataset")
17 print(df)
18
19 print("\nDataset Summary")
20 print(df.describe())
21
22 conversion_rate = (df['Converted'].sum()/len(df))*100
23
24 print(f"\nConversion Rate: {conversion_rate:.2f}%")
25
26 total_sales = df['Sales'].sum()
27 total_spend = df['Marketing_Spend'].sum()
28
29 roi = ((total_sales - total_spend) / total_spend) * 100
30
31 print(f"ROI: {roi:.2f}%")
32
33 df['Age_Group'] = pd.cut(
34     df['Age'],
35     bins=[18,25,35,50],
36     labels=['18-25','26-35','36-50']
37 )
```

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```
38
39 age_sales = df.groupby('Age_Group')['Sales'].sum()
40
41 print("\nSales by Age Group")
42 print(age_sales)
43
44 plt.figure(figsize=(8,5))
45 sns.barplot(x='Customer_ID', y='Sales', data=df)
46 plt.title('Sales by Customer')
47 plt.show()
48
49 plt.figure(figsize=(8,5))
50 sns.scatterplot(
51     x='Marketing_Spend',
52     y='Sales',
53     hue='Gender',
54     data=df,
55     s=100
56 )
57 plt.title('Marketing Spend vs Sales')
58 plt.show()
59
60 age_sales.plot(kind='pie', autopct='%1.1f%%')
61 plt.title('Sales Distribution by Age Group')
62 plt.ylabel('')
63 plt.show()
64
65 print("\nMarketing Analytics Completed Successfully!")
```


Marketing Dataset

	Customer_ID	Age	Gender	Marketing_Spend	Sales	Converted
0	1	22	F	1000	5000	1
1	2	25	M	1500	6000	1
2	3	35	F	2000	9000	1
3	4	45	M	2500	11000	1
4	5	29	F	1800	7500	0
5	6	31	M	2200	9500	1
6	7	40	F	3000	13000	1
7	8	50	M	3500	15000	1
8	9	27	F	1700	7000	0
9	10	38	M	2600	12000	1

Dataset Summary

	Customer_ID	Age	Marketing_Spend	Sales	Converted
count	10.00000	10.000000	10.000000	10.000000	10.000000
mean	5.50000	34.200000	2180.000000	9500.000000	0.800000
std	3.02765	9.052931	742.069179	3231.786572	0.421637
min	1.00000	22.000000	1000.000000	5000.000000	0.000000
25%	3.25000	27.500000	1725.000000	7125.000000	1.000000
50%	5.50000	33.000000	2100.000000	9250.000000	1.000000
75%	7.75000	39.500000	2575.000000	11750.000000	1.000000
max	10.00000	50.000000	3500.000000	15000.000000	1.000000

Conversion Rate: 80.00%

ROI: 335.78%

Sales by Age Group

Age_Group

18-25 11000

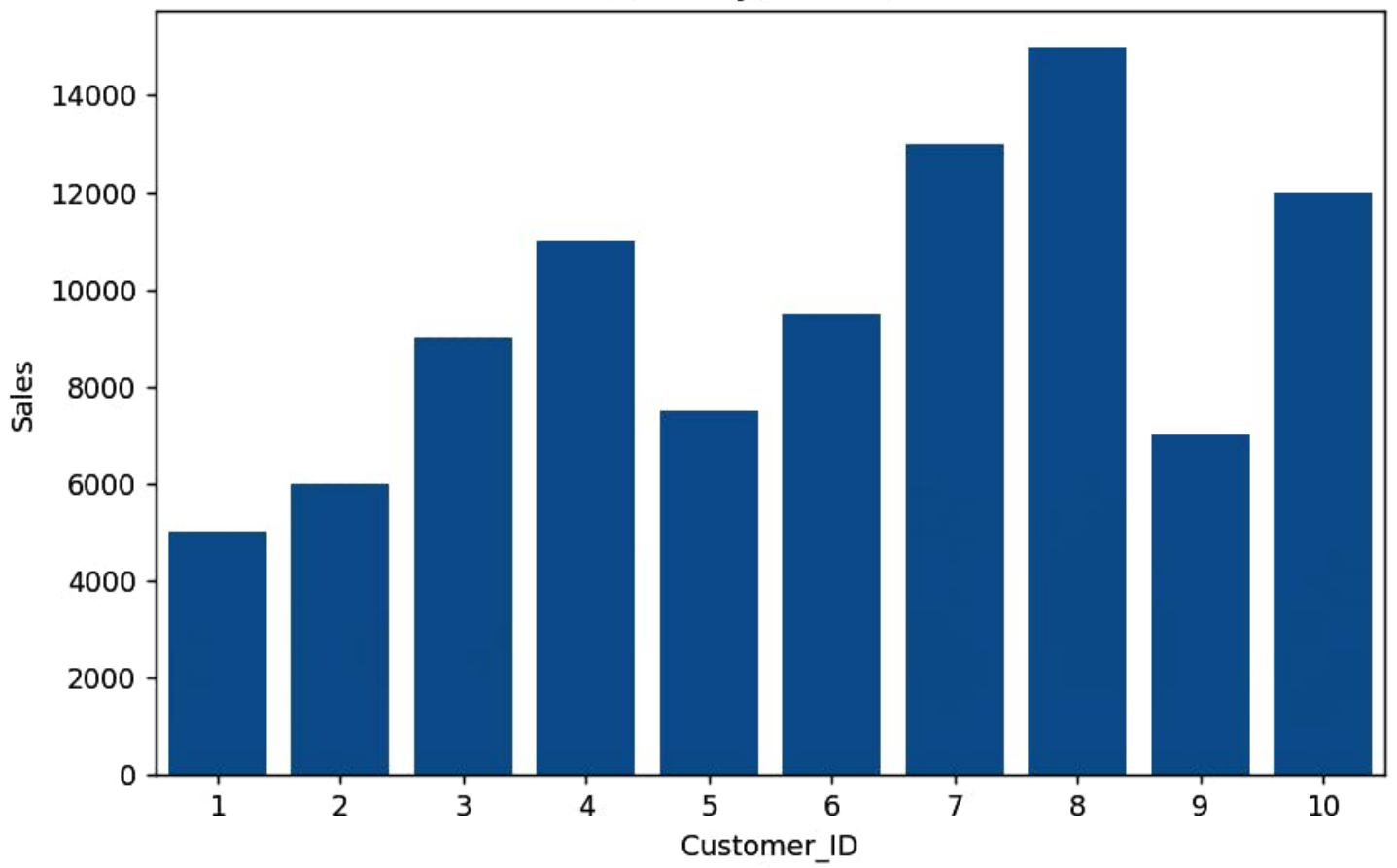
26-35 33000

36-50 51000

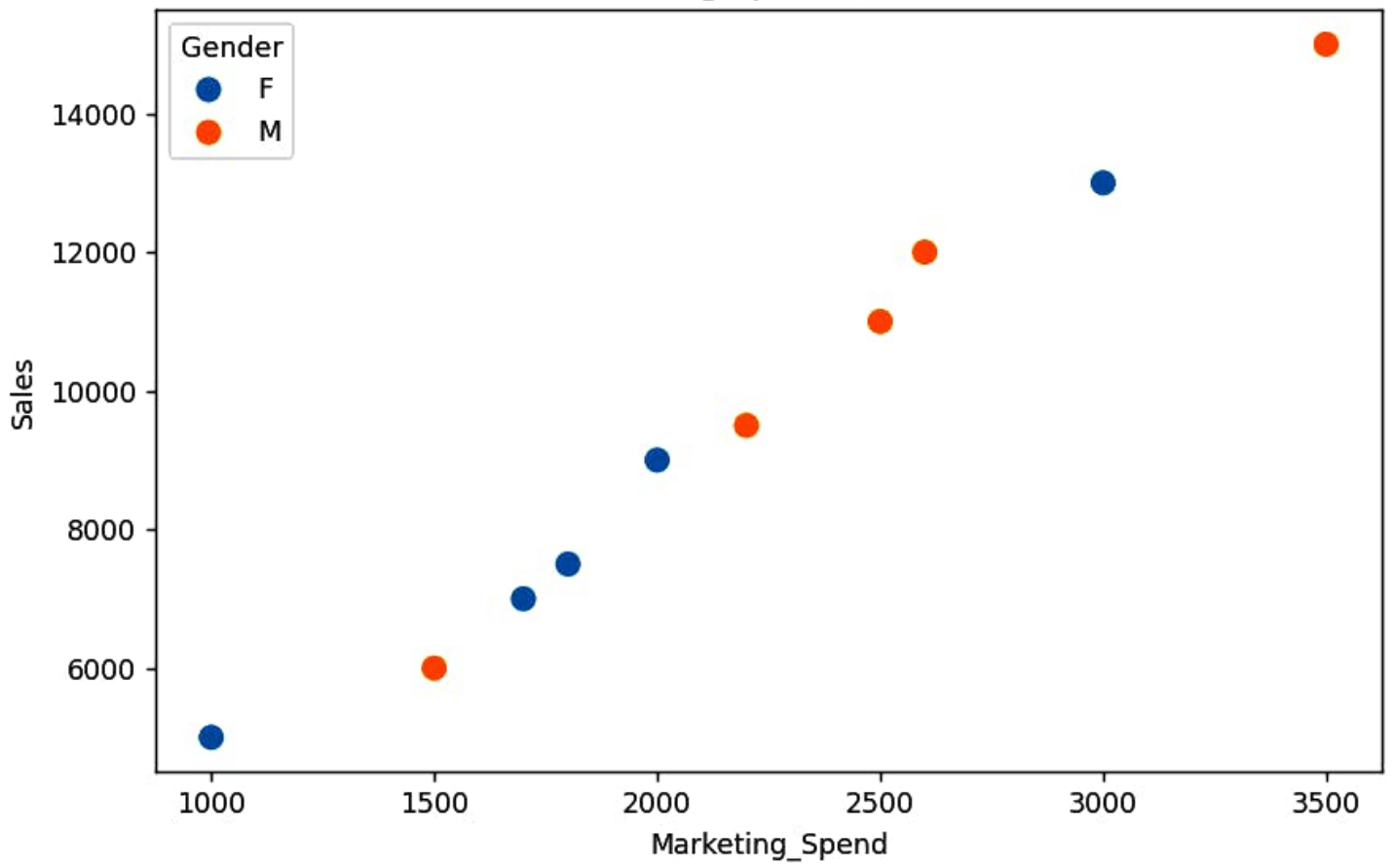
Name: Sales, dtype: int64

Marketing Analytics Completed Successfully!

Sales by Customer



Marketing Spend vs Sales



Sales Distribution by Age Group

